

PANCREATIC SECRETIONS

COMPOSITION AND FUNCTION

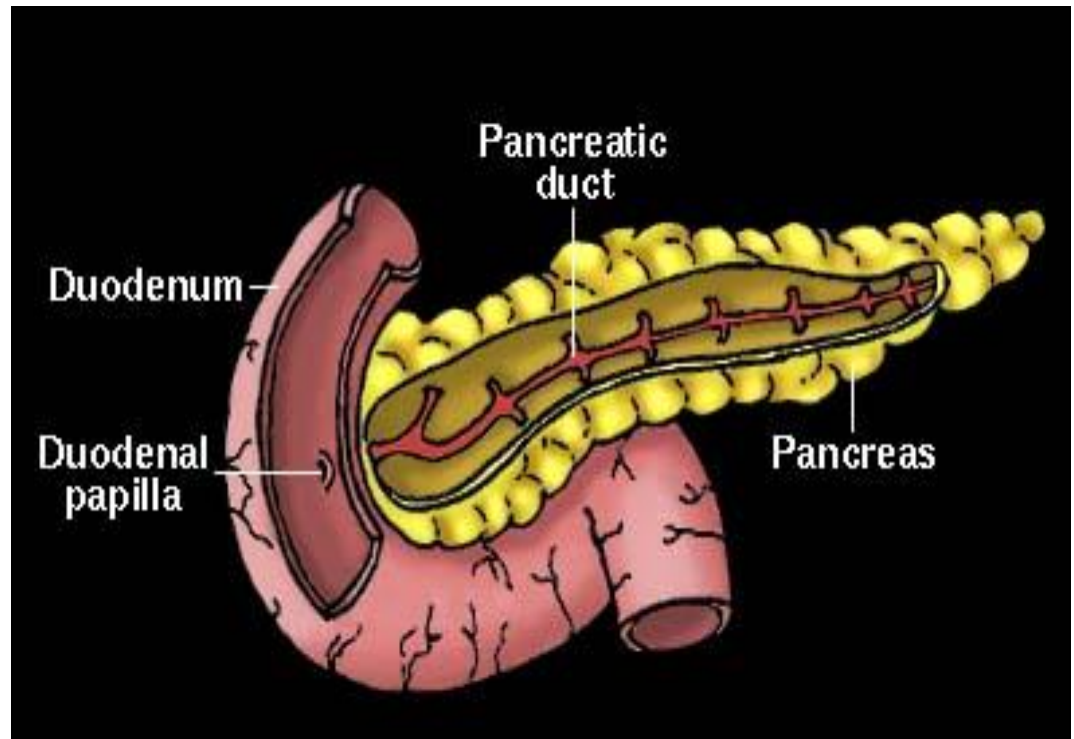
LEARNING *OBJECTIVES*

At the end of lecture students should be able to

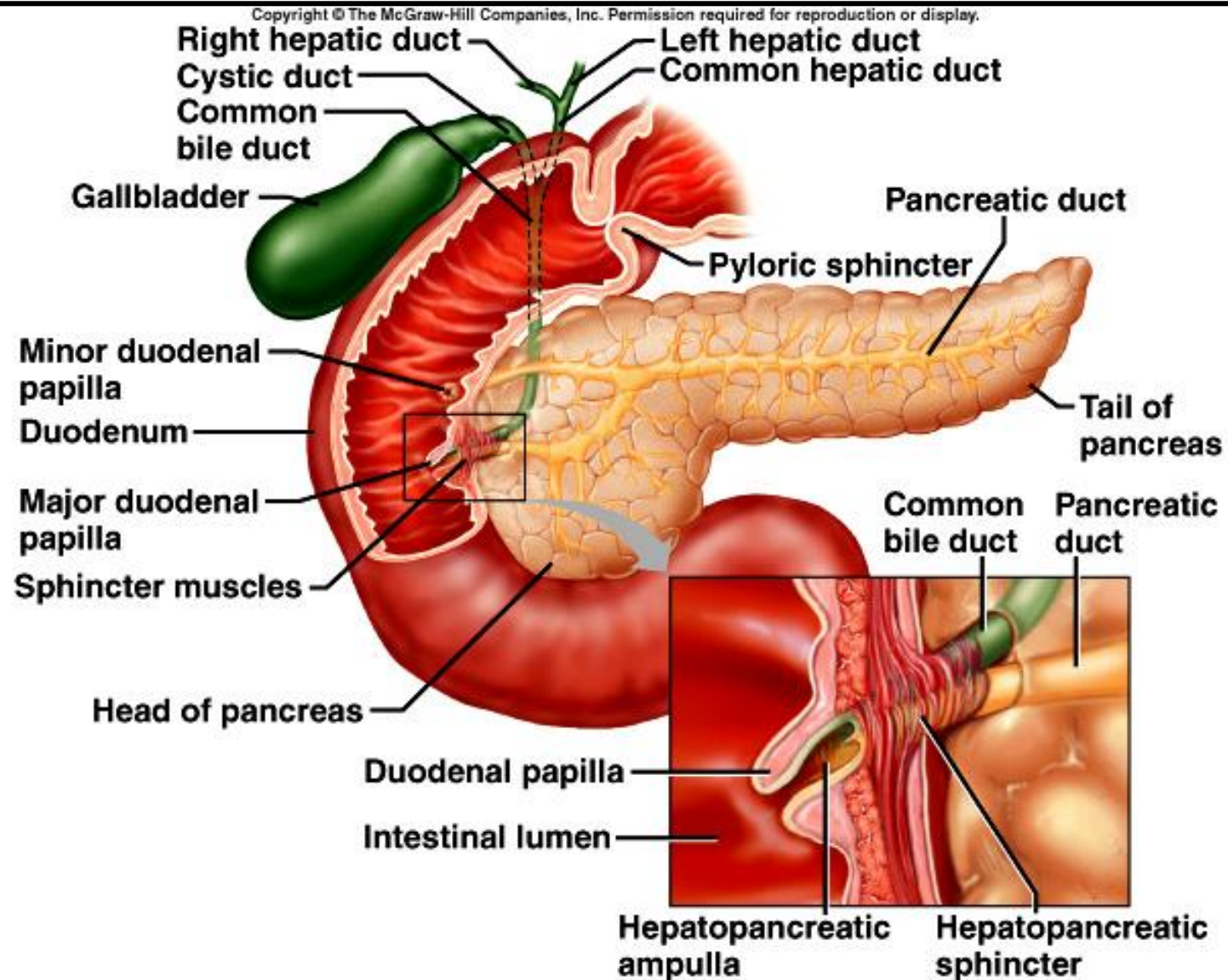
- Describe the structure of pancreas
- List the secretions of pancreas, their cells of origin & their functions
- Differentiate b/w exocrine & endocrine pancreas
- Discuss the regulation of pancreatic secretion
- Illustrate the different phases of pancreatic secretions

SECRETION OF THE PANCREAS

- Endocrine - insulin & glucagon
- Exocrine - enzymes and bicarbonate
- Essential for digestion
- Almost under separate hormonal control

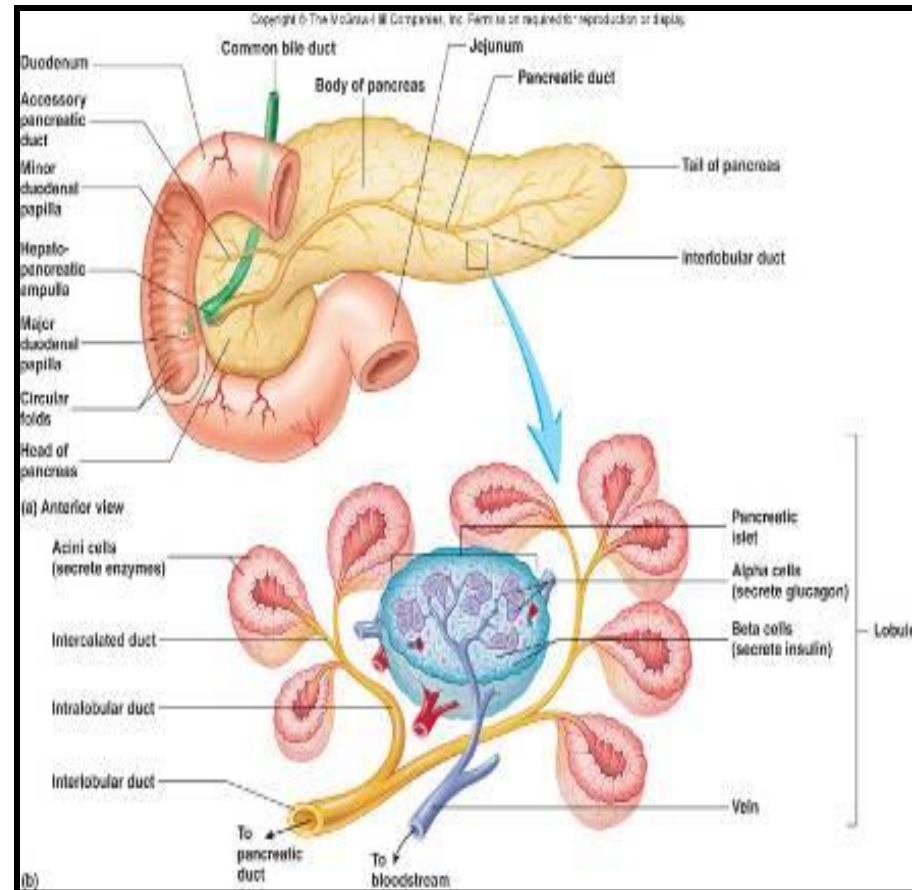


PANCREAS

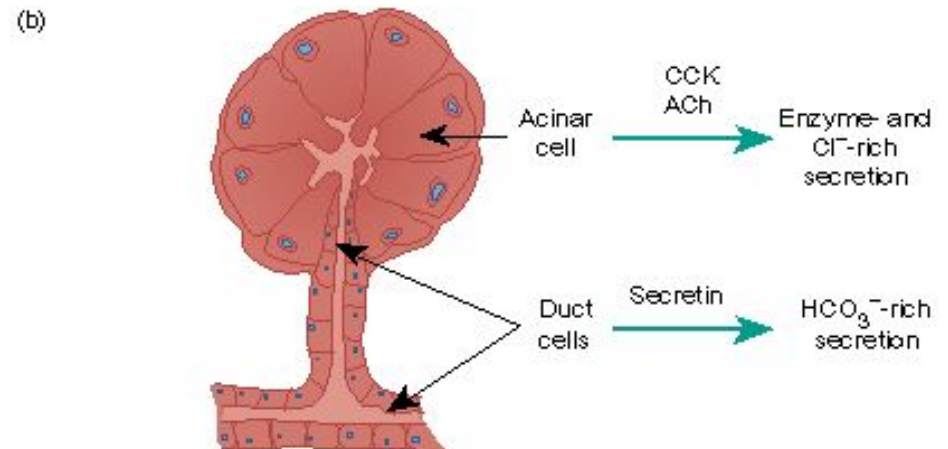
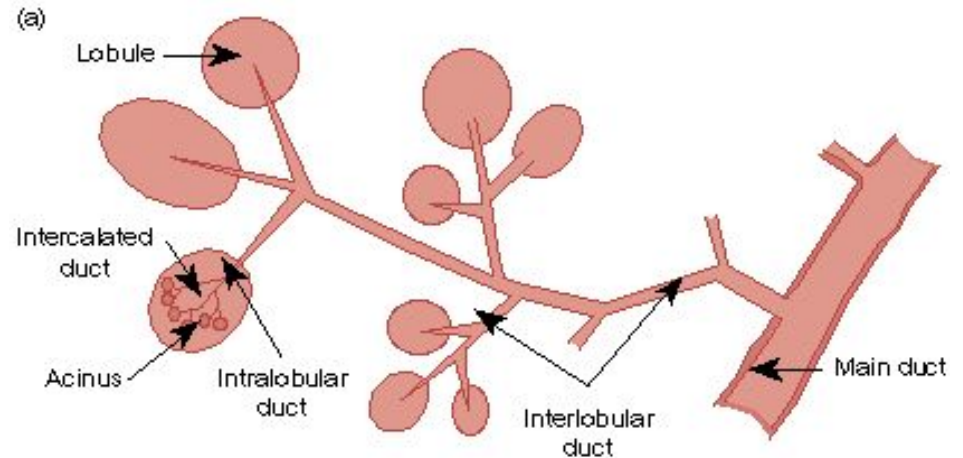
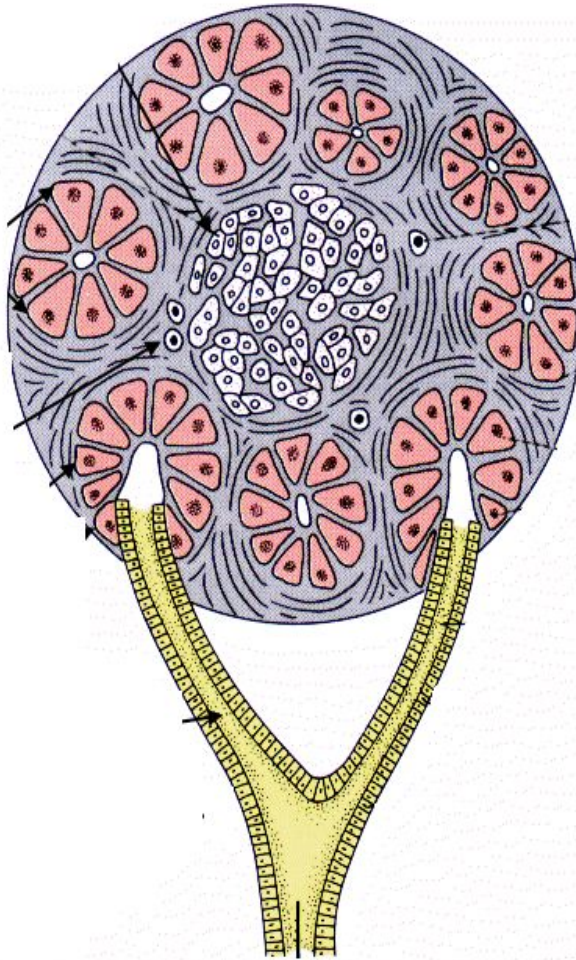


PANCREAS

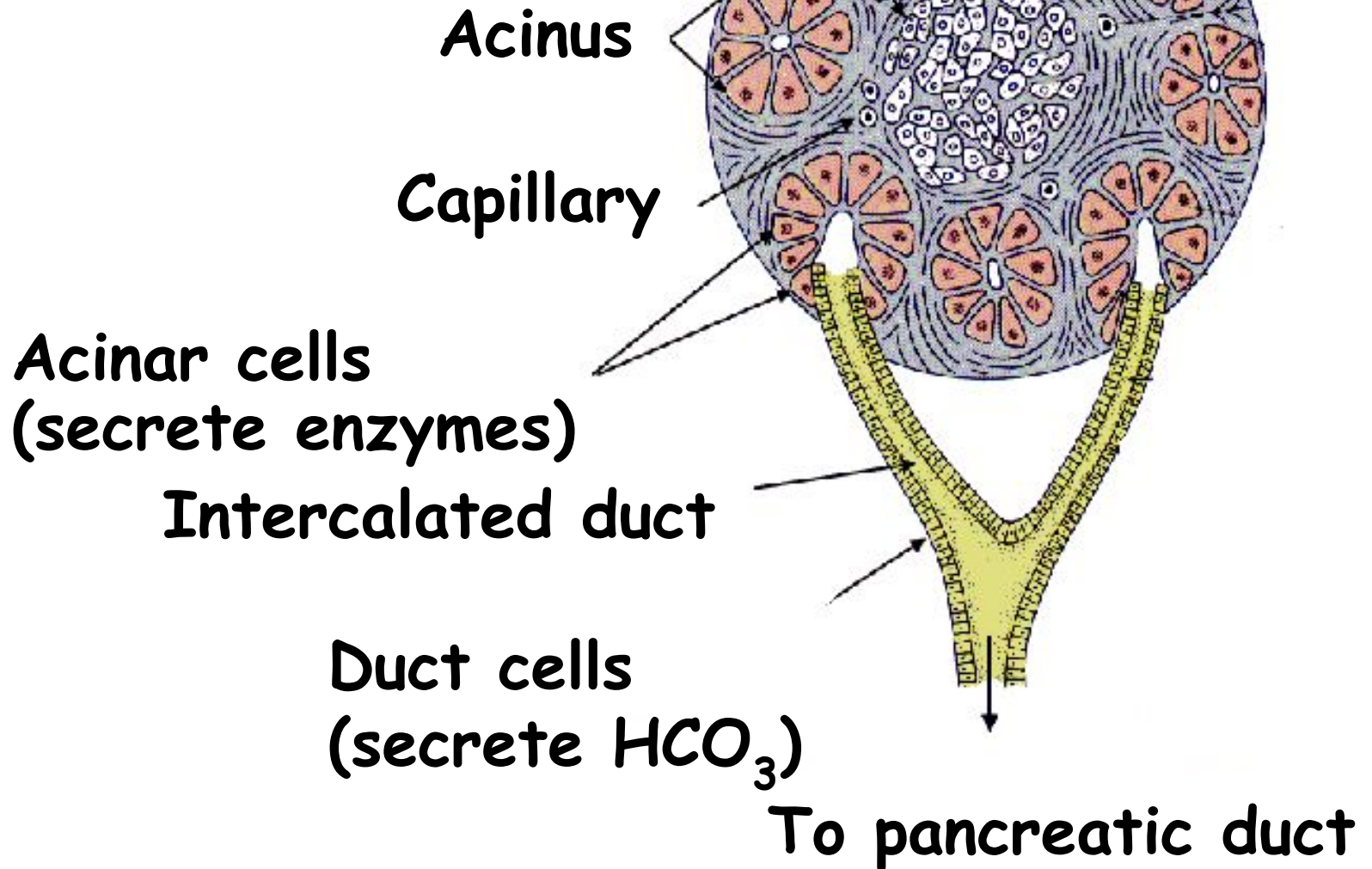
- Pancreas has both endocrine and exocrine parts
- Endocrine: pancreatic islets
Produce insulin, glucagon, and somatostatin
- Exocrine: groups acini (grape-like cluster) form lobules separated by septa.
- Intercalated ducts lead to interlobular ducts which leads to the pancreatic duct
- Pancreatic duct joins common bile duct and enters duodenum at the hepato pancreatic ampulla controlled by the hepatopancreatic ampullar sphincter



ANATOMY AND SECRETION



Islet of Langerhans (secrete insulin)



EXOCRINE PANCREAS

Responsible for digestive function of pancreas

Anatomical Structure

Acini → Ducts → Pancreatic Duct

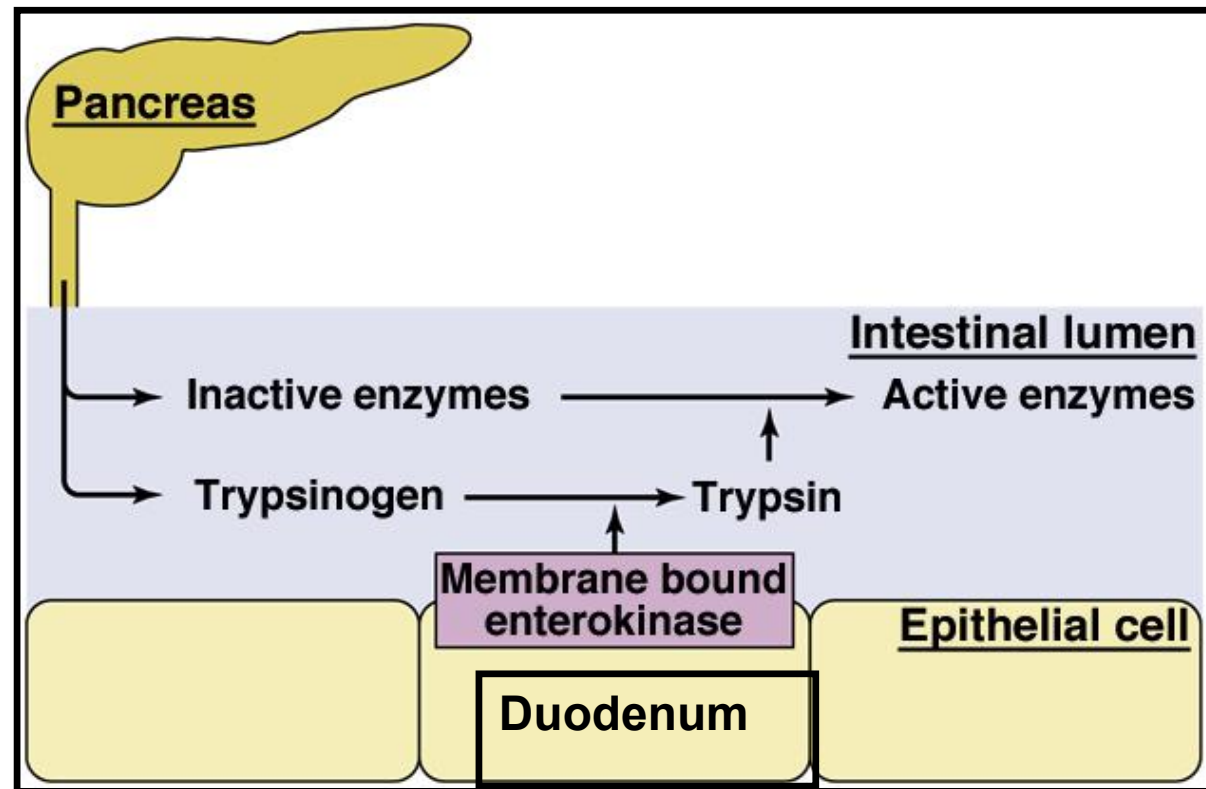
Function

Secretion of bicarbonate by duct cells

Secretion of digestive enzymes by acinar cells

ZYMOGENS

- Acinar cells contain digestive enzymes stored as inactive zymogen granules
- Prevents autodigestion of pancreas
- Enterokinase (bound to brush border of duodenal enterocytes) converts trypsinogen to trypsin
- Trypsin converts all other zymogens to active forms



CATEGORIES OF PANCREATIC ENZYMES

Proteases	Cleave peptide bonds
Nucleases	Hydrolyze DNA/RNA
Elastases	Collagen digestion
Phospholipases	Phospholipids to fatty acids
Lipases	Triglycerides to fatty acids+ glycerol
Amylase	Starch to maltose + glucose

PANCREATIC SECRETIONS: PANCREATIC JUICE

- Aqueous. Produced by columnar epithelium lining smaller ducts. Na^+ , K^+ , HCO_3^- , water. Bicarbonate lowers pH inhibiting pepsin and providing proper pH for enzymes

Enzymatic portion.

- trypsinogen,
- chymotrypsinogen,
- procarboxypeptidase,
- pancreatic amylase,
- pancreatic lipase,
- deoxyribonuclease and ribonuclease.

PANCREATIC SECRETIONS: PANCREATIC JUICE

- Interaction of duodenal and pancreatic enzymes.
 - Enterokinase from the duodenal mucosa and attached to the brush border activates trypsinogen to trypsin.
 - Trypsin activates chymotrypsinogen to chymotrypsin.
 - Trypsin activates procarboxypeptidase to carboxypeptidase.
- Trypsin, chymotrypsin and carboxypeptidase digest proteins: proteolytic.
- Pancreatic amylase continues digestion of starch
- Pancreatic lipase digests lipids
- Deoxyribonucleases and ribonucleases digest DNA and ribonucleic acid, respectively

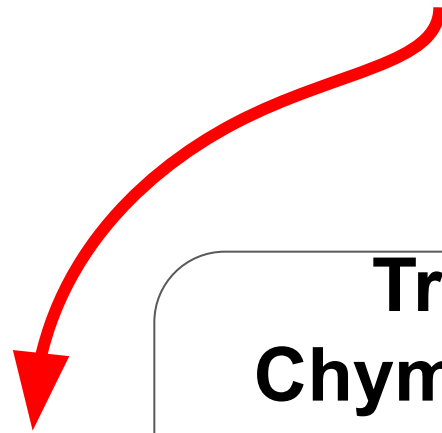
ACTIVATION OF PANCREATIC PROTEASES

Enterokinase

Trypsinogen



Trypsin



Trypsinogen
Chymotrypsinogen
Proelastase
Procarboxypeptidase



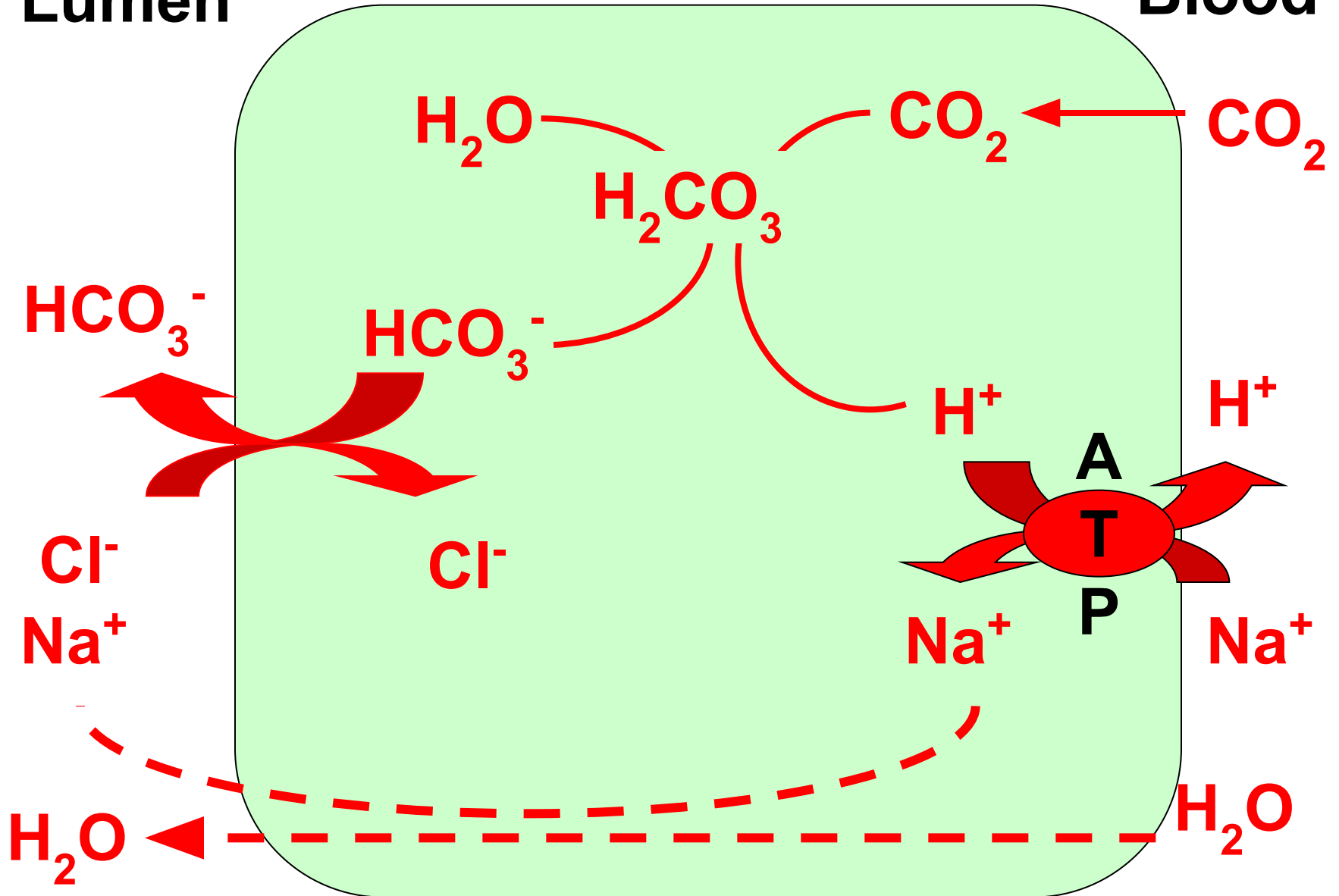
**Trypsin
Chymotrypsin
Elastase
Carboxypeptidase**

- Same pancreatic cells which secrete the proteolytic enzymes, also secrete the trypsin inhibitors, that inhibit TRYPSIN otherwise the normal cells of pancreas would have been digested by trypsin
- When pancreas is damaged or pancreatic duct is blocked, large quantities of pancreatic secretions are pooled and enzymes becomes activated which digest the pancreas within hours(**acute pancreatitis**---leads to shock)

BICARBONATE SECRETION

Lumen

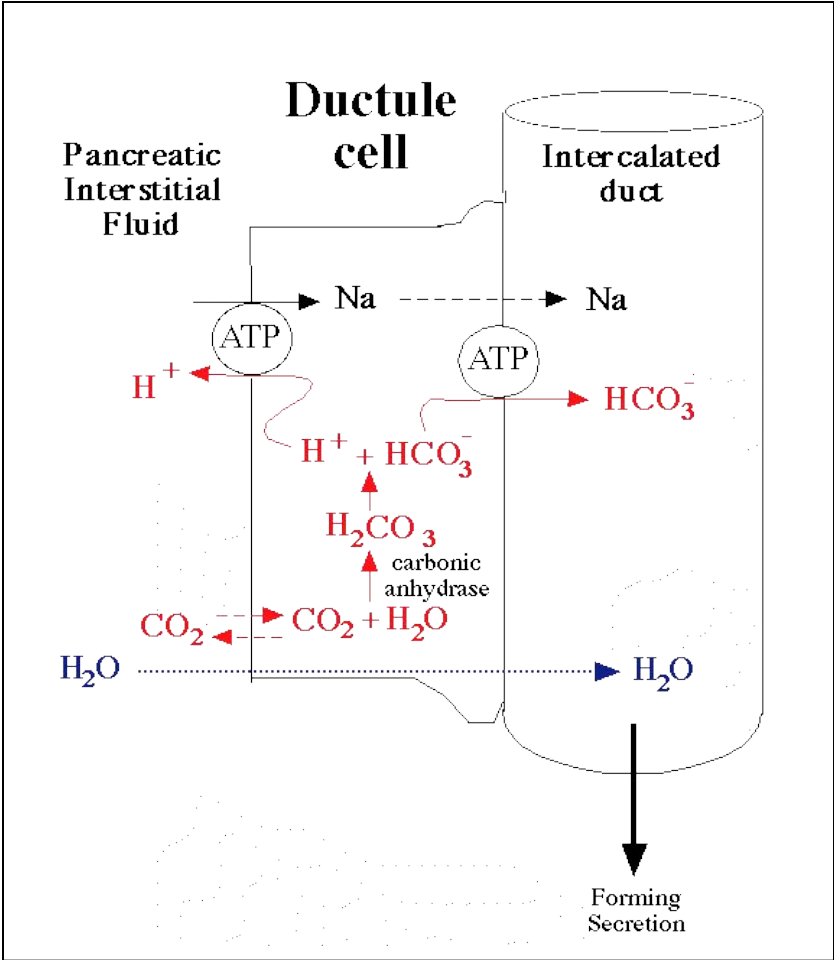
Blood



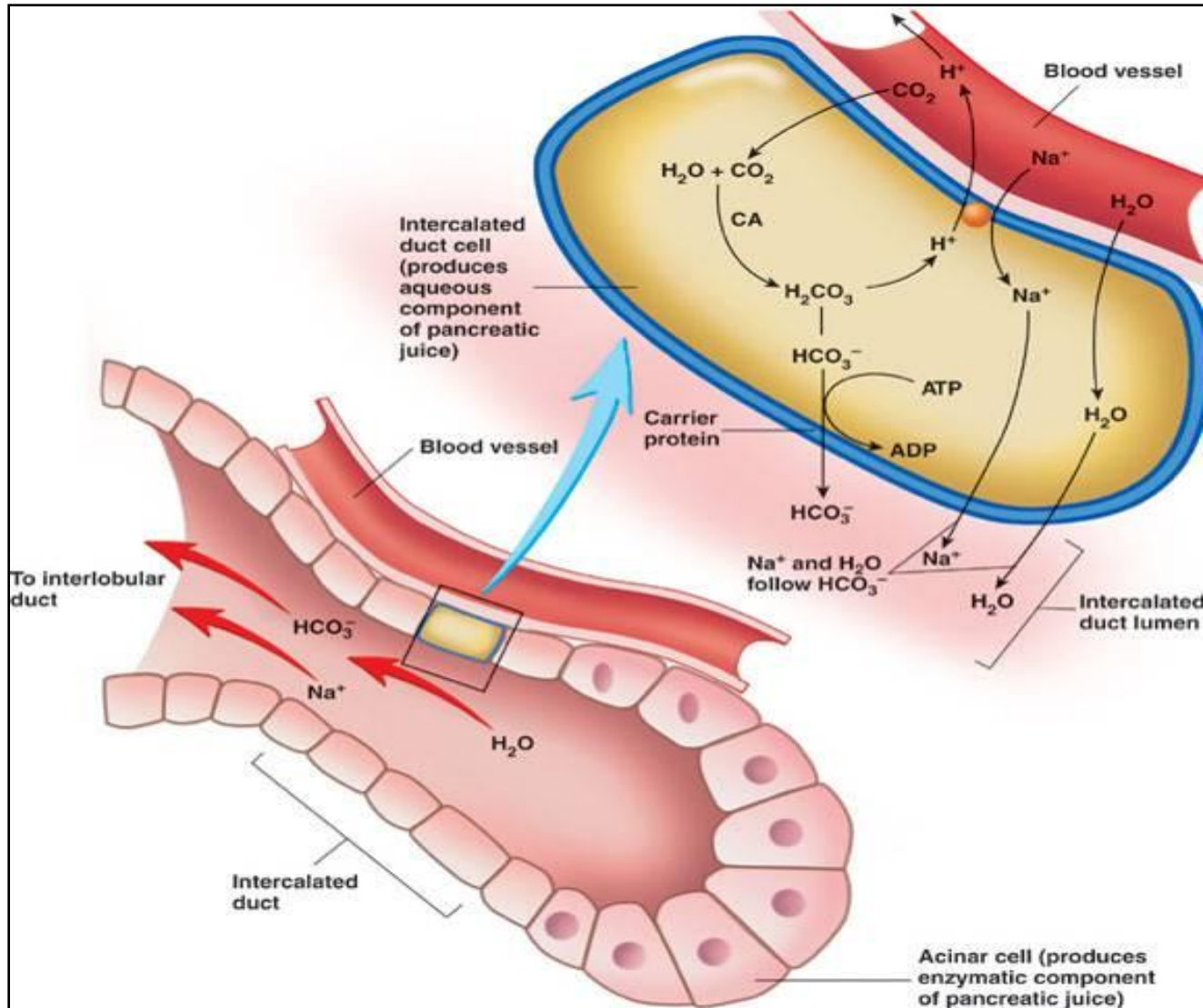
BICARBONATE FUNCTION

Functions

1. Neutralize gastric acid emptied into the duodenum
2. Provide a favorable alkaline environment for optimal activity of pancreatic enzymes



BICARBONATE ION PRODUCTION IN PANCREAS



CONTROL OF PANCREATIC FUNCTION

- **Bicarbonate secretion stimulated by secretin**
- **Secretin released in response to acid in duodenum**
- **Zymogen secretion stimulated by cholecystokinin (CCK)**
- **CCK released in response to fat/amino acids in duodenum**
- **Also under neural control (vagal/local reflexes) - triggered by arrival of organic nutrients in duodenum**

CONTROL OF PANCREATIC SECRETION

Secretion in 3 phases:

Cephalic phase - only 10-15% of total secretion

Activation of vagal efferent stimulates enzyme release

Gastric phase - only present in some species

NOT SIGNIFICANT IN HUMANS

Intestinal phase - majority of secretion

Combination of hormones CCK and secretin and neuronal reflex

Results in maximal enzyme and bicarbonate release

CONTROL OF PANCREATIC SECRETION

Key hormones in stimulation of secretion are:

Cholecystokinin (CCK) and **Secretin** (released from the small intestine)

Inhibiting factors: SS, PP, glucagon

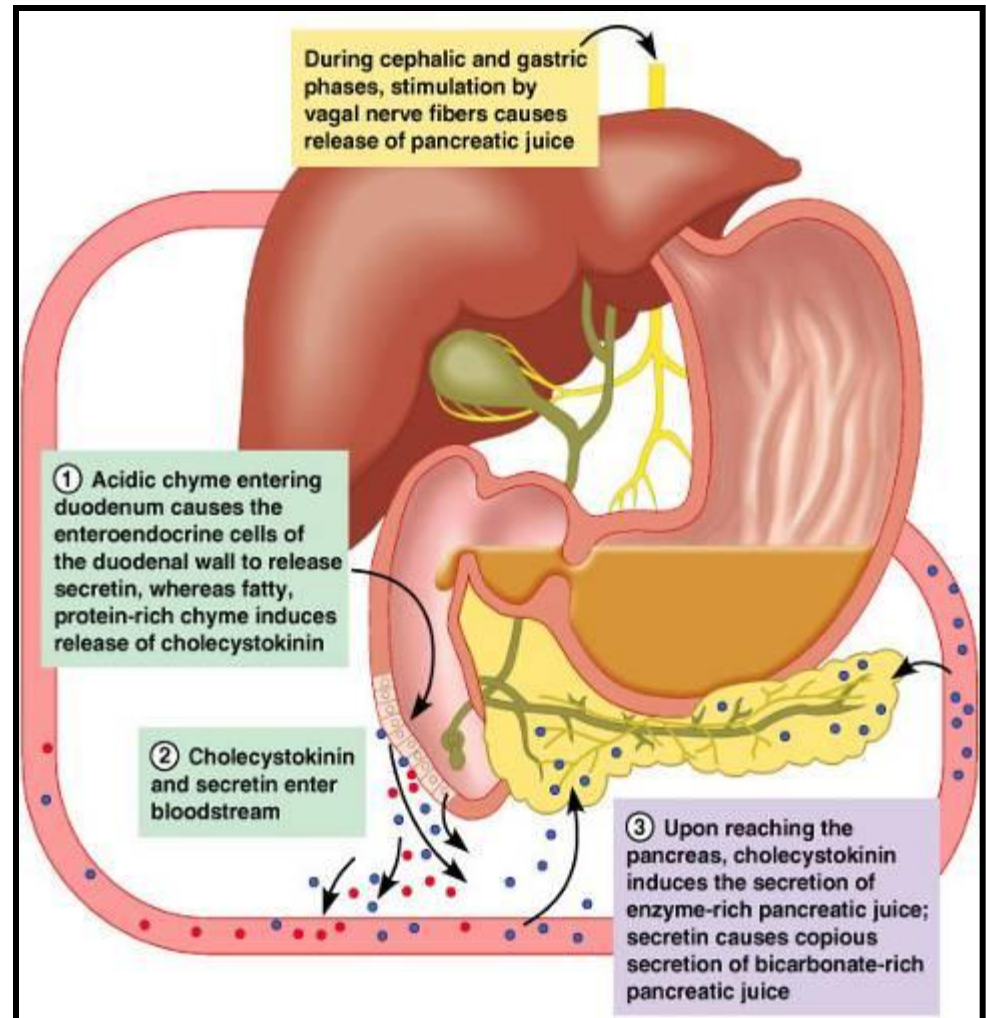
REGULATION OF PANCREATIC SECRETION

Stimuli:

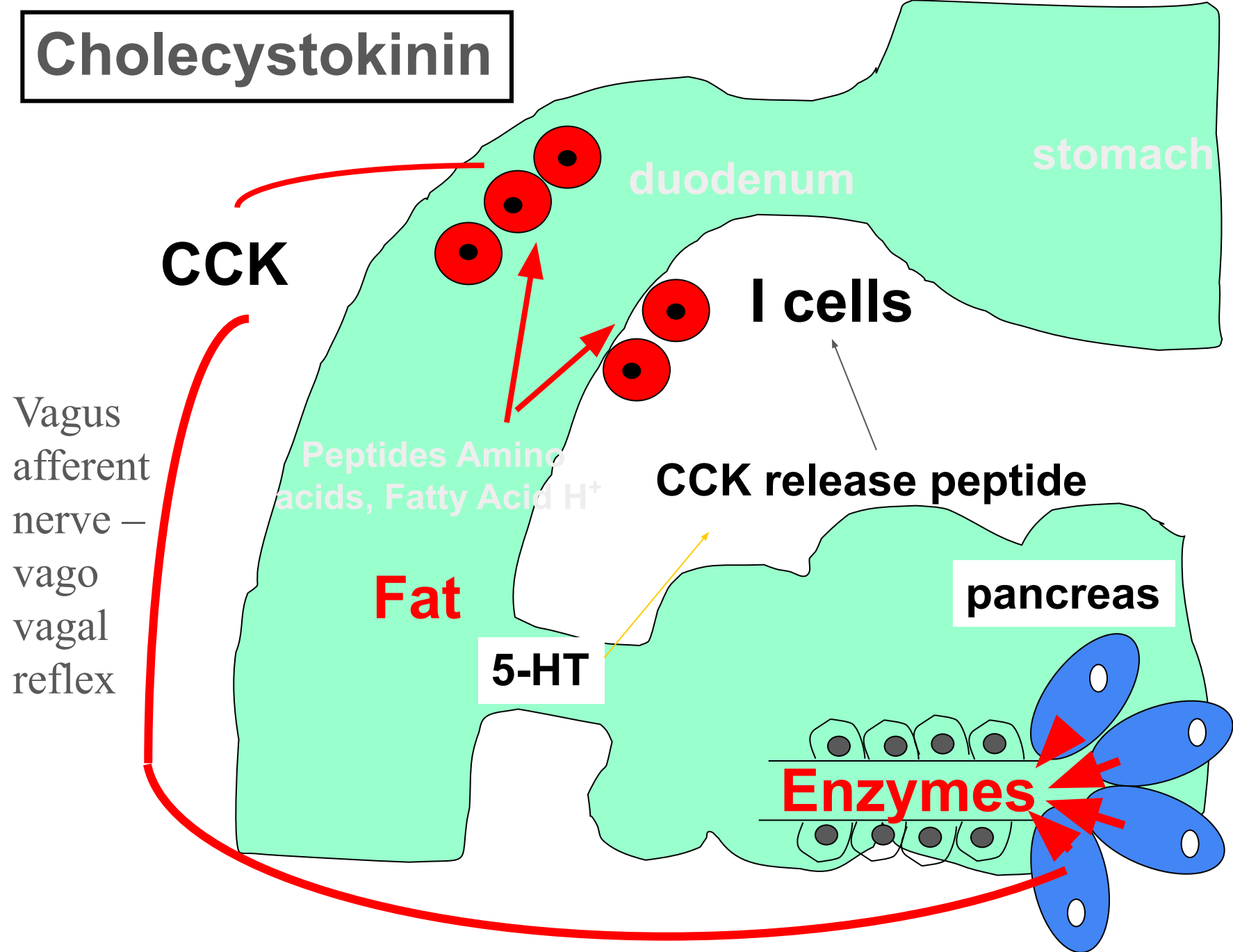
1 **Acetyl choline**
(vagus n. ending)

2 **CCK**

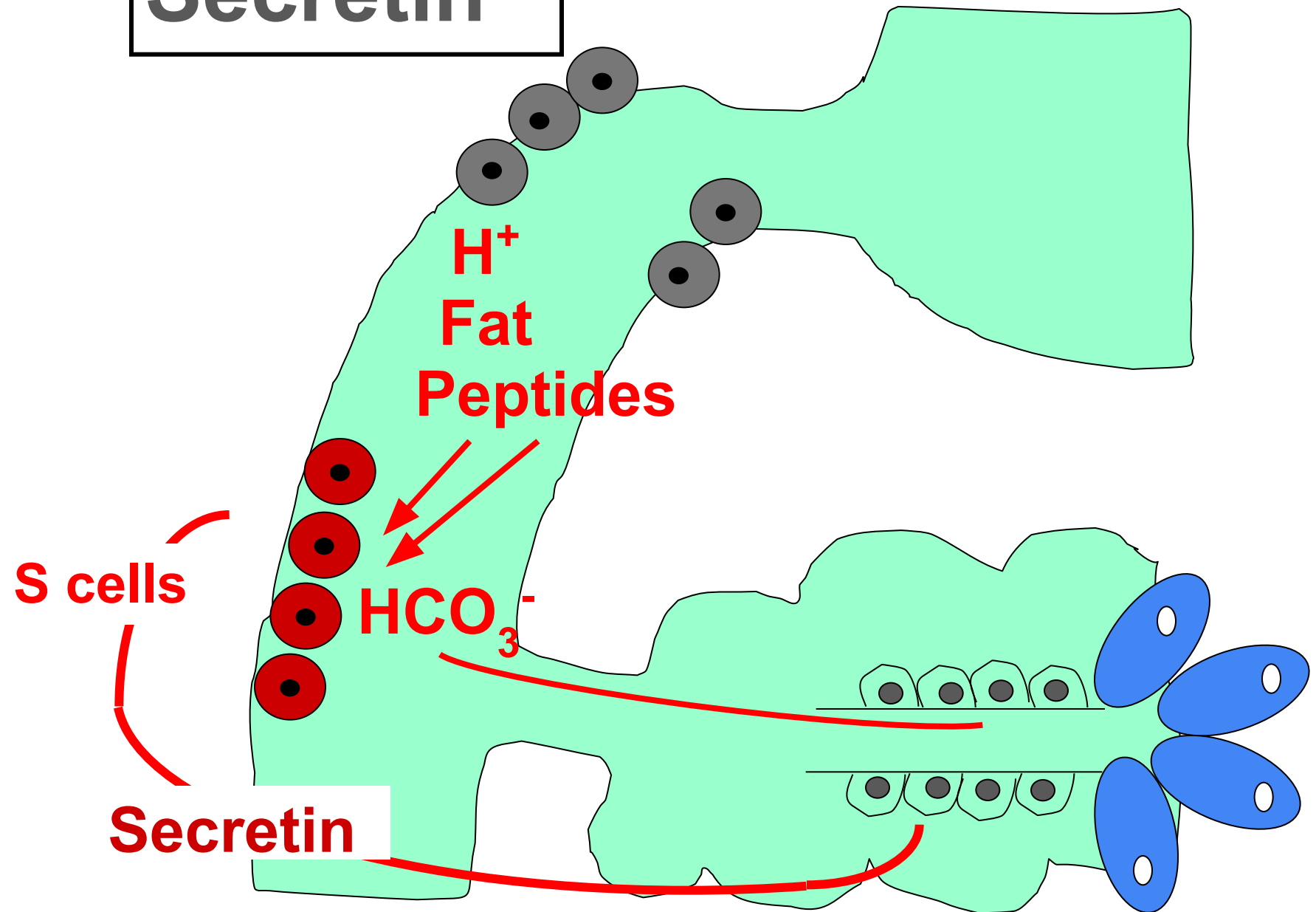
3 **Secretin**



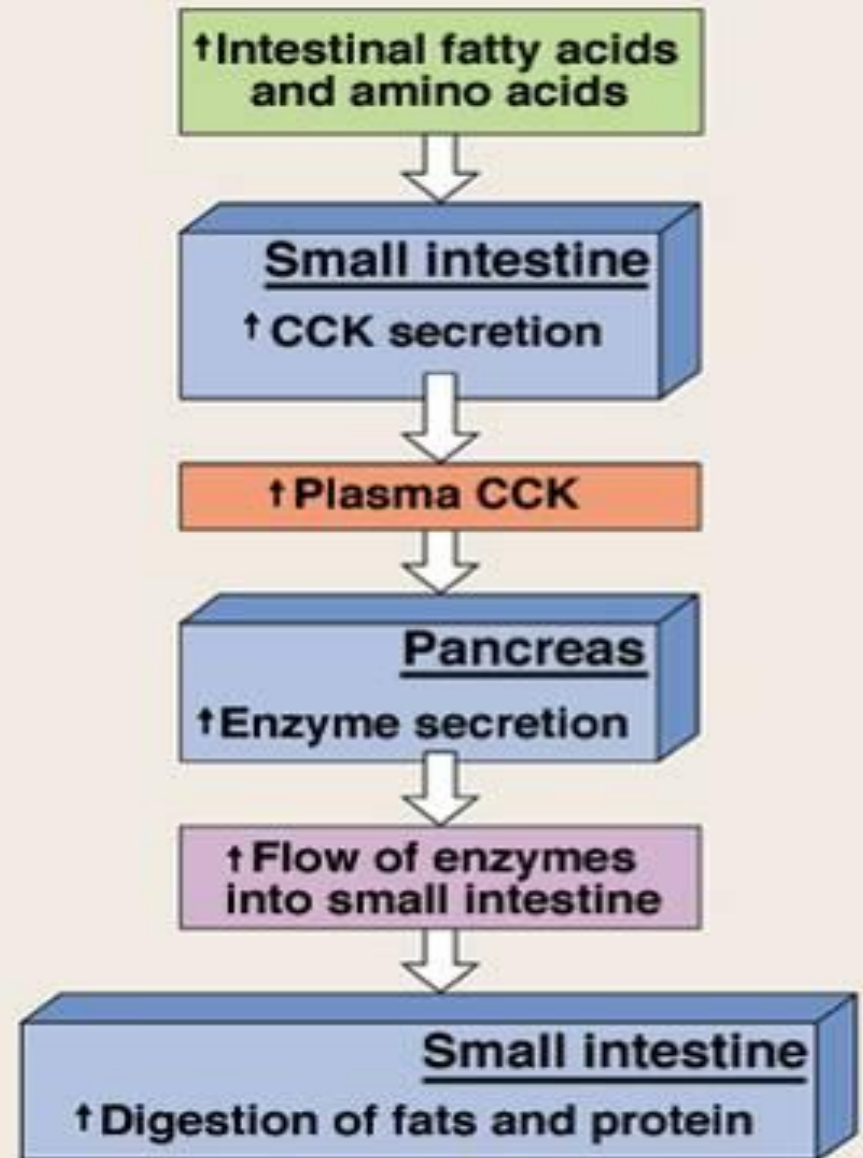
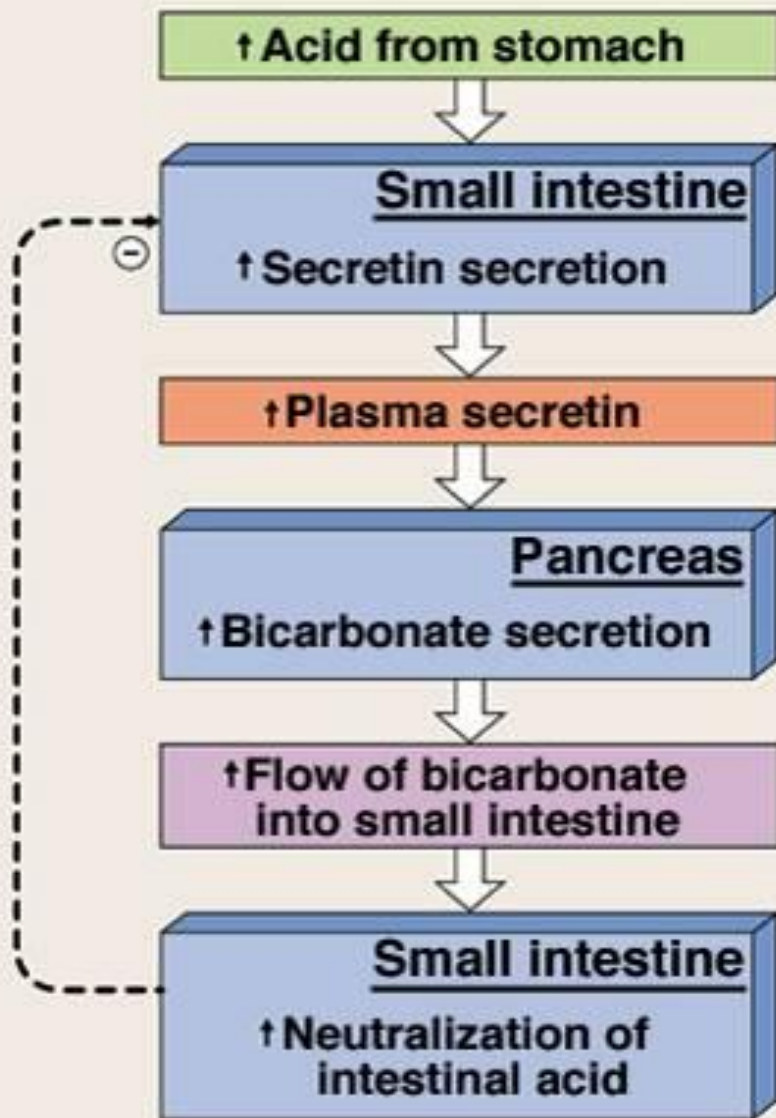
Cholecystokinin



Secretin



CONTROL OF PANCREATIC FUNCTION



INTESTINAL PHASE OF SECRETION

